ESO208 COMPUTATIONAL METHODS IN ENGINEERING

ASSIGNMENT 8

MAXIMUM MARKS: 30

DUE DATE: 11 OCTOBER 2026

Both questions have to be solved by hand. Please upload hand-written solutions on Gradescope as a single PDF file.

Q.1. [10 marks] The following data points of a function $f(x)$ are given.

x	0.5	1.0	1.5	2.0
$f(x)$	0.9197	0.6767	0.3734	0.1831

Calculate the derivative $f'(x)$ at $x = 1$ using:

(a) Forward divided difference of the order O(h). [1.5]

(b) Forward divided difference of the order O(h^2). [1.5]

(c) Backward divided difference of the order O(h). [1.5]

(d) Central divided difference of the order O(h^2). [1.5]

(e) If $f(x) = 5xe^{-2x}$, calculate the errors in (a)-(d). Which estimate in (a)-(d) is the most accurate? Explain your answer using the graph of $f'(x)$ [4]

Q.2. [20 marks] The respiratory quotient is a dimensionless number used to estimate the metabolic rate of microorganisms. It is computed as the amount of carbon dioxide produced per unit oxygen consumed by microorganisms. In addition, the rates can be integrated to determine the total amounts of carbon dioxide produced and oxygen consumed over the course of the experiment. The following cumulative quantities of the two gases were measured during an experiment.

Time (hr)	0	1	2	3	4	5	6	7	8	9	10
CO_2 generated (g)	0	75	151	225	297	367	436	502	568	634	701
O_2 consumed (g)	0	76	149	224	301	381	463	547	631	715	801

(a) Use numerical differentiation to determine the respiratory quotient for each time-step. Use forward divided difference of the order O(h^2) at t=0, backward divided difference of the order O(h^2) at t=10 and central divided difference of the order O(h^2) at t = 1, 2, 3 ..... 9. [10]

(b) Numerically integrate the results in (a) to calculate the amount of CO_2 generated at t=10 using the trapezoidal rule. [5]

(c) Numerically integrate the results in (a) to calculate the amount of CO_2 generated at t=10 using Simpson's 1/3 rule. [5]